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Week-3

Step 1: Verify the book_meta data collection

The screenshot displays the MongoDB Atlas Data Explorer interface. The left sidebar shows the 'Data Explorer' section with a tree view of clusters. The 'book_metadata' collection is selected under the 'bookflow' cluster. The main panel shows the 'Documents' tab with a list of documents. The first document is for 'The Martian' by Andy Weir, published in 2021, and the second is for 'Dune' by Frank Herbert, published in 2022. The interface includes a search bar, a query builder, and a 'Find' button. The bottom status bar indicates 'Alert System Status: All Good'.

ORGANIZATION: Tejaswini's Org - 202... PROJECT: Project 0

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

25 1 - 5 of 5

```
{
  "_id": ObjectId("6a7eb6c8dd8ed99e6ffc9b97e"),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3)
}
```

```
{
  "_id": ObjectId("6a7eb6e0dd8ed99e6ffc9b988"),
  "book_id": 102,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "3.1 MB"
}
```

Alert System Status: All Good
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Step 2: Comparison Query Using \$gt, \$in

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdata

customers

transactions

bookflow

book_metadata

local

sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 6AggregationsSchemaIndexes 1Validation

{

"reviews.rating": {

"\$gt": 4

}

}

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a7eb6c8dd8ed99e6ffcb97e')

book_id: 181

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 181

reviews: Array (3)

_id: ObjectId('6a7eb6e8dd8ed99e6ffcb988')

book_id: 182

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 182

reviews: Array (3)

_id: ObjectId('6a7eb6f8dd8ed99e6ffcb982')

book_id: 183

title: "Clean Code"

author: "Robert C. Martin"

published_year: 2019

category: "Programming"

format: "Physical"

dimensions: Object (3)

member_id: 183

reviews: Array (2)

_id: ObjectId('6a7eb783dd8ed99e6ffcb984')

book_id: 184

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 184

reviews: Array (3)

_id: ObjectId('6a7eb714dd8ed99e6ffcb986')

book_id: 185

title: "Data Structures"

author: "Mark Allen"

published_year: 2024

category: "Computer Science"

format: "Digital PDF"

file_size: "5.8 MB"

member_id: 185

reviews: Array (2)



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Project 0



Data Explorer



{ } My Queries



🔗 Data Modeling



CLUSTERS (1)



Cluster0

admin

bankdb

customers

transactions

bookflow

book_metadata

local

sample_mflix

book_metadata



Cluster0 > bookflow > book_metadata

Documents (5)

Aggregations

Schema

Indexes (1)

Validation



```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

+ ADD DATA

BULK

EXPORT CODE



```
_id: ObjectId('6a7eb6c8dd0ed99e6ffcb97e')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
  0: Object (3)
  1: Object (3)
  2: Object (3)
```

```
_id: ObjectId('6a7eb6e0dd0ed99e6ffcb980')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
category: "Science Fiction"
format: "Digital EPUB"
file_size: "3.1 MB"
member_id: 102
reviews: Array (3)
```

```
_id: ObjectId('6a7eb6f0dd0ed99e6ffcb982')
book_id: 103
title: "Clean Code"
author: "Robert C. Martin"
published_year: 2019
category: "Programming"
format: "Physical"
dimensions: Object (3)
member_id: 103
reviews: Array (2)
```

```
_id: ObjectId('6a7eb703dd0ed99e6ffcb984')
book_id: 104
title: "Artificial Intelligence Basics"
author: "John Smith"
published_year: 2023
category: "Artificial Intelligence"
format: "Digital PDF"
file_size: "4.2 MB"
member_id: 104
reviews: Array (3)
```

```
_id: ObjectId('6a7eb714dd0ed99e6ffcb986')
book_id: 105
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 105
reviews: Array (3)
```

Step 3: Regex Query

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Data Explorer

{ } My Queries

🔗 Data Modeling

CLUSTERS (1)

Cluster0

▶ admin

▶ bankdb

customers

transactions

▶ bookflow

book_metadata

▶ local

▶ sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

{
 "format": {
 "\$regex": "Digital",
 "\$options": "i"
 }
}

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a7eb6c8dd0ed99e6ffcb97e')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
▶ reviews: Array (3)

_id: ObjectId('6a7eb6e9dd0ed99e6ffcb980')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
category: "Science Fiction"
format: "Digital EPUB"
file_size: "3.1 MB"
member_id: 102
▶ reviews: Array (3)

_id: ObjectId('6a7eb703dd0ed99e6ffcb984')
book_id: 104
title: "Artificial Intelligence Basics"
author: "John Smith"
published_year: 2023
category: "Artificial Intelligence"
format: "Digital PDF"
file_size: "4.2 MB"
member_id: 104
▶ reviews: Array (3)

_id: ObjectId('6a7eb714dd0ed99e6ffcb986')

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CLUSTERS (1)

Search clusters

Cluster0

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transactions

bookflow

book_metadata

local

sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

{

"format": {

"\$regex": "Digital",

"\$options": "i"

}

}

ADD DATA

BULK

EXPORT CODE

reviews : Array (3)

_id: ObjectId('6a7eb6e0dd0ed99e6ffcb980')

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Array (3)

_id: ObjectId('6a7eb703dd0ed99e6ffcb984')

book_id: 104

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 104

reviews: Array (3)

_id: ObjectId('6a7eb714dd0ed99e6ffcb986')

book_id: 105

title: "Data Structures"

author: "Mark Allen"

published_year: 2024

category: "Computer Science"

format: "Digital PDF"

file_size: "5.0 MB"

member_id: 105

reviews: Array (2)

Step 4: Aggregation Pipeline

The screenshot displays the MongoDB Atlas Data Explorer interface for a cluster named 'bookflow' in the 'book_metadata' collection. The interface shows the aggregation pipeline being built, with three stages: \$match, \$unwind, and \$group.

Stage 1: \$match

```
1 //
2 // query: The query in MQL
3 //
4 {
5   "published_year": {
6     "$gt": 2020
7   }
8 }
```

Output preview after \$match stage (Sample of 4 documents):

Document 1	Document 2	Document 3	Document 4
{ "_id": "6a7b6c8d8b9d9e9fcb07e", "book_id": 181, "title": "The Martian", "author": "Andy Weir", "published_year": 2011, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "72.5 MB", "average_rating": 4.5 } ... 17 C more ...	{ "_id": "6a7b6c8d8b9d9e9fcb080", "book_id": 181, "title": "Dune", "author": "Frank Herbert", "published_year": 2002, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "72.5 MB", "average_rating": 4.5 } ... 17 C more ...	{ "_id": "6a7b6c8d8b9d9e9fcb082", "book_id": 181, "title": "Clean Code", "author": "Robert C. Martin", "published_year": 2019, "category": "Programming", "format": "Physical", "file_size": "72.5 MB", "average_rating": 4.5 } ... 17 C more ...	{ "_id": "6a7b6c8d8b9d9e9fcb084", "book_id": 181, "title": "Artificial Intelligence Basics", "author": "John Smith", "published_year": 2023, "category": "Artificial Intelligence", "format": "Digital PDF", "file_size": "72.5 MB", "average_rating": 4.5 } ... 17 C more ...

Stage 2: \$unwind

```
1 {
2   "path": "$reviews"
3 }
```

Output preview after \$unwind stage (Sample of 10 documents):

Document 1	Document 2	Document 3	Document 4
{ "_id": "6a7b6c8d8b9d9e9fcb07e", "book_id": 181, "title": "The Martian", "author": "Andy Weir", "published_year": 2011, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "72.5 MB", "average_rating": 4.5, "reviews": [{ "rating": 5, "text": "Great book!" }] } ... 17 C more ...	{ "_id": "6a7b6c8d8b9d9e9fcb07e", "book_id": 181, "title": "The Martian", "author": "Andy Weir", "published_year": 2011, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "72.5 MB", "average_rating": 4.5, "reviews": [{ "rating": 4, "text": "Good book!" }] } ... 17 C more ...	{ "_id": "6a7b6c8d8b9d9e9fcb07e", "book_id": 181, "title": "The Martian", "author": "Andy Weir", "published_year": 2011, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "72.5 MB", "average_rating": 4.5, "reviews": [{ "rating": 3, "text": "Average book!" }] } ... 17 C more ...	{ "_id": "6a7b6c8d8b9d9e9fcb07e", "book_id": 181, "title": "The Martian", "author": "Andy Weir", "published_year": 2011, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "72.5 MB", "average_rating": 4.5, "reviews": [{ "rating": 2, "text": "Not great!" }] } ... 17 C more ...

Stage 3: \$group

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents):

Document 1	Document 2	Document 3	Document 4
{ "_id": "Dune", "average_rating": 4.666666666666667, "total_reviews": 3 } ... 17 C more ...	{ "_id": "Data Structures", "average_rating": 4, "total_reviews": 2 } ... 17 C more ...	{ "_id": "The Martian", "average_rating": 4.666666666666667, "total_reviews": 3 } ... 17 C more ...	{ "_id": "Artificial Intelligence Basics", "average_rating": 4.333333333, "total_reviews": 3 } ... 17 C more ...

At the bottom of the interface, there is a status bar indicating 'All System Status: All Good' and a copyright notice for MongoDB, Inc. The interface also includes a sidebar with 'Data Explorer' and 'My Queries' sections, and a top navigation bar with 'Compare tiers' and 'Visualize Your Data' options.

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Project 0

Compare Sets

View monitoring

Visualize Your Data

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- banktab
 - customers
 - transactions
- bookflow
 - book_metadata
- local
- sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Match

Sumwind

Group

Sort

Generate aggregation Explain Run Options

match - modified SAVE CREATE NEW EXPORT CODE

Stage 3 \$group

Output preview after \$group if stage (Sample of 4 documents)

_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Data Structures"
average_rating: 4
total_reviews: 2

_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Artificial Intelligence"
average_rating: 4.333333333333333
total_reviews: 3

Stage 4 \$sort

Output preview after \$sort if stage (Sample of 4 documents)

_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Artificial Intelligence Basics"
average_rating: 4.333333333333333
total_reviews: 3

_id: "Data Structures"
average_rating: 4
total_reviews: 2

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Compare Sets

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Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- banktab
 - customers
 - transactions
- bookflow
 - book_metadata
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- sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Match

Sumwind

Group

Sort

Generate aggregation Explain Run Options

match - modified SAVE CREATE NEW EXPORT CODE

```
1 * {
2 *   $match:
3 *     /**
4 *      * query: The query in MQL.
5 *      */
6 *   {
7 *     published_year: {
8 *       $gt: 2020
9 *     }
10 *   },
11 *   $unwind: {
12 *     path: "$reviews"
13 *   },
14 *   $group: {
15 *     _id: "$title",
16 *     average_rating: {
17 *       $avg: "$reviews.rating"
18 *     },
19 *     total_reviews: {
20 *       $sum: 1
21 *     }
22 *   },
23 *   $sort: {
24 *     average_rating: -1
25 *   }
26 * }
```

PIPELINE OUTPUT PREVIEW

Sample of 4 documents

_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3

_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3

_id: "Artificial Intelligence Basics"
average_rating: 4.333333333333333
total_reviews: 3

_id: "Data Structures"
average_rating: 4
total_reviews: 2